Modelling censored log-normal responses in generalized additive models

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This document details the technical issues surrounding the implementation of censored log-normal responses in the R package `mgcv`. Necessary background information is provided along with derivations necessary for computation.

1 log-normal modelling

What does it mean to model log-normal data? What we are saying is that we have some variable, Y, which is $\log_b$-normally[1] distributed so $Y \sim \log_b N(\mu, \sigma^2)$ where μ is the logarithm of the mean and σ^2 is the base b logarithm of the variance of Y. The log-normal distribution is useful in cases where Y is defined over the interval $(0, \infty]$ (i.e., $Y > 0$).

Canonical references for the log-normal distribution include Aitchison and Brown (1957) and Crow and Shimizu (1988).

1.1 Arithmetic versus geometric means

Starting out by thinking only about the natural logarithm case, we can estimate the expectation of $\mathbb{E}(Y)$ by either an *arithmetic* or *geometric* mean (denoted $\mathbb{E}_A(Y)$ or $\mathbb{E}_G(Y)$, respectively). When we use a natural logarithm (e), these are defined as:

$$\begin{aligned} \mathbb{E}_A(Y) &= e^{\mu + \frac{1}{2}\sigma^2} \quad \text{and} \\ \mathbb{E}_G(Y) &= e^{\mu}. \end{aligned} \tag{1}$$

Where the geometric "mean" corresponds to the median of the distribution. This difference is shown in Figure 1.

Similarly, we define the arithmetic and geometric standard deviations as follows:

$$\begin{aligned} \mathrm{SD}_A(Y) &= e^{\mu + \frac{1}{2}\sigma^2}\sqrt{e^{\sigma^2} - 1} \\ \mathrm{SD}_G(Y) &= e^{\sigma}. \end{aligned}$$

In the more general case where we have a base b logarithm, we have that:

$$\begin{aligned} \mathbb{E}_A(Y) &= b^{\mu + \frac{1}{2}\sigma^2 \log_e b} \quad \text{and} \\ \mathbb{E}_G(Y) &= b^{\mu}. \end{aligned} \tag{2}$$

and

$$\begin{aligned} \mathrm{SD}_A(Y) &= b^{\mu + \frac{1}{2}\sigma^2 \log_e b}\sqrt{b^{\sigma^2 \log_e b} - 1} \\ \mathrm{SD}_G(Y) &= b^{\sigma}. \end{aligned}$$

[1] Note that throughout I use $\log_a$ to indicate logarithm to the base a since these calculations work regardless of the base used. When the distribution is referred to informally (i.e., non-mathematically) I drop the base.

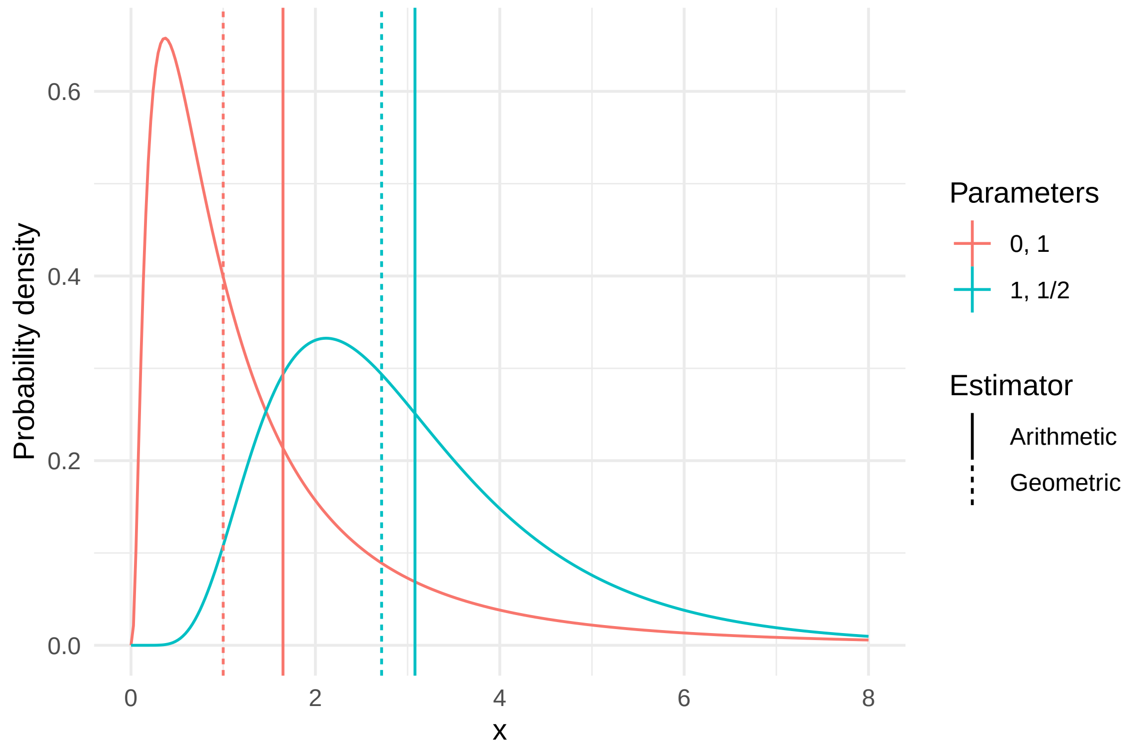

Figure 1: Demonstration of the differences in estimation of arithmetic vs geometric means for log-normal, overlaid on the distribution functions. Parameters are given on the $\log_e$ scale, as required by the R function `dlnorm`.

Following usual results, we have that multiplication of log-normal distributed variables results in a log-normally distributed product and therefore geometric means of log-normally distributed variables are log-normally distributed. We also have a multiplicative central limit theorem, where geometric means of (any) random variables are approximately log-normally distributed (under usual conditions).

1.2 log-normal response in regression-like models

We may want to model the mean via some linear model, for example when using a GLM or GAM[2]. We decompose the mean as $\mathbf{X}\boldsymbol{\beta}$, where $\mathbf{X}$ is an $n \times p$ design matrix and $\boldsymbol{\beta}$ is a p-vector of coefficients to be estimated (where n is the number of observations).

In practice, one can simply assume that the log-response is normally distributed and fit a GLM-a-like[3] with a normal distribution. This model is:

$$\mathbb{E}(Y) = b^{\mu} = b^{\mathbf{X}\boldsymbol{\beta}},$$

which is clearly modelling the geometric, rather than arithmetic mean (equivalently, we are modelling the median). This naturally extends to the case of random effects and therefore penalized smoothers (i.e., GLMMs, GAMs and GAMMs).

An example of such an analysis in R is as follows:

```
library(mgcv)

## Loading required package:  nlme
## This is mgcv 1.9-4.  For overview type '?mgcv'.
```

[2]Basic theory for log-normal linear models is covered in Section 3.2 of Crow and Shimizu (1988).
[3]GAM/GLMM/GAMM/etc.

```
set.seed(2)

# generate some simple data
n <- 10000
x <- rnorm(n, sd=0.05)
beta0 <- 1
beta1 <- 1.2
mu <- beta0+beta1*x
s <- 0.2

# generate response
z <- rlnorm(n, meanlog=mu, sdlog=s)

# fit model
aa <- gam(log(z)~x, method="REML")
summary(aa)
##
## Family: gaussian
## Link function: identity
##
## Formula:
## log(z) ~ x
##
## Parametric coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 0.999854   0.002024  493.92   <2e-16 ***
## x           1.255988   0.040499   31.01   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## R-sq.(adj) =  0.0877   Deviance explained = 8.78%
## -REML = -1778.1  Scale est. = 0.040974  n = 10000
# compare RSS
sum((z-exp(predict(aa)))^2)
## [1] 3274.107
sum((z-exp(predict(aa)+1/2*aa$sig2))^2)
## [1] 3242.197
```

1.3 Bias correction for random effects models

As noted by Thorson and Kristensen (2016) (and references therein) when random effects are used alongside non-linear transformations such as logarithms, we can incur a bias due to that transformation. This correction is of the order of the inverse of the sample size, so will become negligible at larger sample sizes. Since splines are just highly structured random effects (Miller, 2025), we need to ensure that we take this bias correction into account. This can be especially important when making aggregate predictions on the response scale (for example when summing over areas) and when the response distribution is skewed (one of the primary use cases of the log-normal distribution).

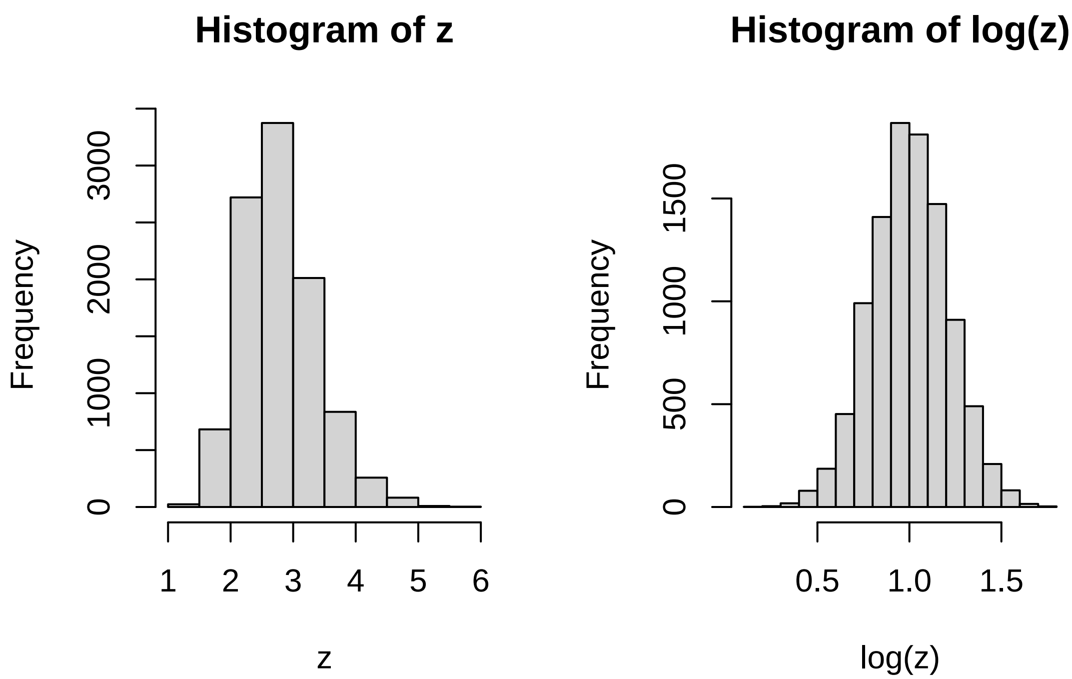

Figure 2: Histograms of the data generated in the linear model example. Left showing the data and right showing it log-transformed, looking more normal.

2 Censored normal modelling

Censored normal models are one kind of tobit model[4] (Goldberger, 1964) where the response is censored in some way (in our case, that observations are below some detection threshold). Core to their formulation is the separation of the observation into censored and non-censored groups, where the censored likelihood contributions are evaluations of the cumulative distribution function (CDF) of the normal distribution.

Data takes the form of a triplet: $(y, \mathbf{x}, c)$ where y is the response, $\mathbf{x}$ is the corresponding covariate vector (or in the case of a GAM, vector of basis function evaluations), and c is an indicator which is 1 if y was censored at some point C (i.e., $y \leq C$) and 0 otherwise. Since we do not know the value of y when $y \leq C$, we need to use the cumulative density function to evaluate the likelihood up to C. For $y > C$ we can use the usual likelihood function, since we then know y.

For modelling censored normal variables, we conveniently know a lot about the CDF of the normal distribution; evaluation is therefore very simple and can use built-in functions in R.

2.1 Implementation in `mgcv` via `cnorm()`

From the `mgcv` help file `?cnorm`: "If y is the response with mean m and standard deviation $w^{-1/2}\exp(\theta)$, then $w^{1/2}(y-m)\exp(-\theta)$ follows an $N(0,1)$ distribution. That is $y \sim N(m, \exp(2\theta)/w)$. θ is a single scalar for all observations. Observations may be left, interval or right censored or uncensored."

[4]"Tobin's probit" tends to refer to any censored regression model.

3 Combining approaches: censored log-normal distribution with arithmetic means for GAMs

We can combine the above approaches to allow for censored log-normal predictions of the arithmetic mean using a GAM while also correcting for bias incurred by the random effects. As such, we have three components to our model:

1. `cnorm()`-based fitting of the model on the log-normally distributed data;
2. correction to use the arithmetic mean, following (2);
3. bias correction from the random effects per Thorson and Kristensen (2016) derived in the Appendix of this document.

We can apply 2 and 3 via *post hoc* corrections to the predictions. These corrections lead to the following expressions:

$$\mathbb{E}\left[g\left(\hat{\boldsymbol{\beta}}\right)\right] = g\left(\hat{\boldsymbol{\beta}}\right) + \frac{1}{2n}\text{trace}\left(D^2 g(\hat{\boldsymbol{\beta}})\Sigma\right) \quad \text{and}$$

$$\hat{\text{Var}}\left[g\left(\hat{\boldsymbol{\beta}}\right)\right] = (Dg\left(\hat{\boldsymbol{\beta}}\right))^T\Sigma(Dg\left(\hat{\boldsymbol{\beta}}\right))/n = \frac{1}{2n^2}\text{trace}\Bigg(\left[\Sigma\left(D^3h(\hat{\boldsymbol{\beta}})\Sigma Dg(\hat{\boldsymbol{\beta}}) - D^2 g(\hat{\boldsymbol{\beta}})\right)\right]^2 \tag{3}$$

$$- \Sigma\left(D^4h(\hat{\boldsymbol{\beta}})\Sigma Dg(\hat{\boldsymbol{\beta}})\Sigma Dg(\hat{\boldsymbol{\beta}}) + D^3h(\hat{\boldsymbol{\beta}})\left[\Sigma\left(2D^2g(\hat{\boldsymbol{\beta}})\Sigma Dg(\hat{\boldsymbol{\beta}}) - D^3h(\hat{\boldsymbol{\beta}})\Sigma Dg(\hat{\boldsymbol{\beta}})\Sigma Dg(\hat{\boldsymbol{\beta}})\right)\right] - 2D^3g(\hat{\boldsymbol{\beta}})\Sigma Dg(\hat{\boldsymbol{\beta}})\right)\Bigg) \tag{4}$$

where $g(\boldsymbol{\beta}) = b^{X\boldsymbol{\beta}+\frac{1}{2}\sigma^2 \log_e b}$ (the arithmetic mean expression). Then b is the base we are working in and X is the design matrix for our predictions. D indicates the derivative operator with respect to $\boldsymbol{\beta}$, with powers of D giving that order of derivative. $\mathbf{V}_{\boldsymbol{\beta}} = \Sigma/n$ is the posterior covariance matrix for $\boldsymbol{\beta}$ ($\mathbf{V}_{\boldsymbol{\beta}}$ is provided by `mgcv::vcov`; see, e.g. Miller (2025) for further explanation of covariance matrices in the context of empirical Bayes modelling).

Derivations for these expressions are given in the Appendix.

4 Implementation of `mgcv` family: `clognorm()`

As the corrections in (3) can be applied post hoc, the easiest mechanism is via `mgcv`'s per-family prediction functions. Since version 1.8-0, it has been possible to implement a `$predict` function within a family (response distribution) function. When `mgcv::predict()` is used with the `type=''response''` flag set this function is called instead allowing for per-family special prediction functions. It is this mechanism that we exploit to implement seamless predictions using base `mgcv` functions.

The `clognorm` family is therefore implemented in `mgcvUtils` as a copy of the `cnorm` library with the addition of a `$predict` method using (3). We allow an extra specification of the base to be used when calling the function during modelling (e.g., `base=10` for $\log_{10}$ or `base=''e''` for $\log_e$).

4.1 A quick example of usage

```
library(mgcv)
library(mgcvUtils)
library(ggplot2)

# make a 1D example
set.seed(123)

# simulated example from gam
data <- gamSim(1, n = 400, dist = "normal", scale = 1)
```

```
## Gu & Wahba 4 term additive model
noise <- rnorm(nrow(data), sd = 0.1)
data$y2 <- 10^(data$f2 / 10 + noise)

# censor values below some value
data$y2 <- cbind(data$y2, data$y2)

# indicate which values are censored, see ?cnorm for rules
cen_level <- 0.5
data$y2[data$y2[, 1] < cen_level, 2] <- -Inf
data$y2[data$y2[, 1] < cen_level, 1] <- cen_level

b2 <- gam(y2~s(x2), data=data, method="REML", family=clognorm(base=10))
summary(b2)
##
## Family: clog10norm(0.698)
## Link function: identity
##
## Formula:
## y2 ~ s(x2)
##
## Parametric coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  2.74700    0.03489   78.74   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##         edf Ref.df Chi.sq p-value
## s(x2) 8.916  8.998   3162  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.888   Deviance explained =   89%
## -REML = 451.23  Scale est. = 1         n = 400
```

We can compare the predictions from the two estimators in Figure (3).

Appendix

"Thorson's" epsilon correction

The core of the REML/ML methods implemented in `mgcv` is Laplace-approximate maximum likelihood (LAML). In the case where Laplace approximations are used, we can use the correction "proposed" by Thorson and Kristensen (2016) and elaborated on in Thorson and Kristensen (2024)[5]. Scare quotes are used here as the correction is based on Tierney et al. (1989), from their Theorem 2. They propose estimators for the expectation and variance of some function g of the model parameters ($\hat{\theta}$) when a model is estimated using a Laplace approximation.

We can generally write the expectation as:

$$\mathbb{E}\left[g(\theta)\right] = \frac{\int g(\theta)L(\theta)\pi(\theta)d\theta}{\int L(\theta)\pi(\theta)d\theta}. \tag{5}$$

[5]Note that there are a couple of mathematical errors in both of these sources.

```
## Warning in transformation$transform(x):  NaNs produced
## Warning in scale_y_continuous(trans = "log10"):  log-10 transformation introduced infinite
values.
## Warning:  Removed 1 row containing missing values or values outside the scale range
## (`geom_ribbon()`).
```

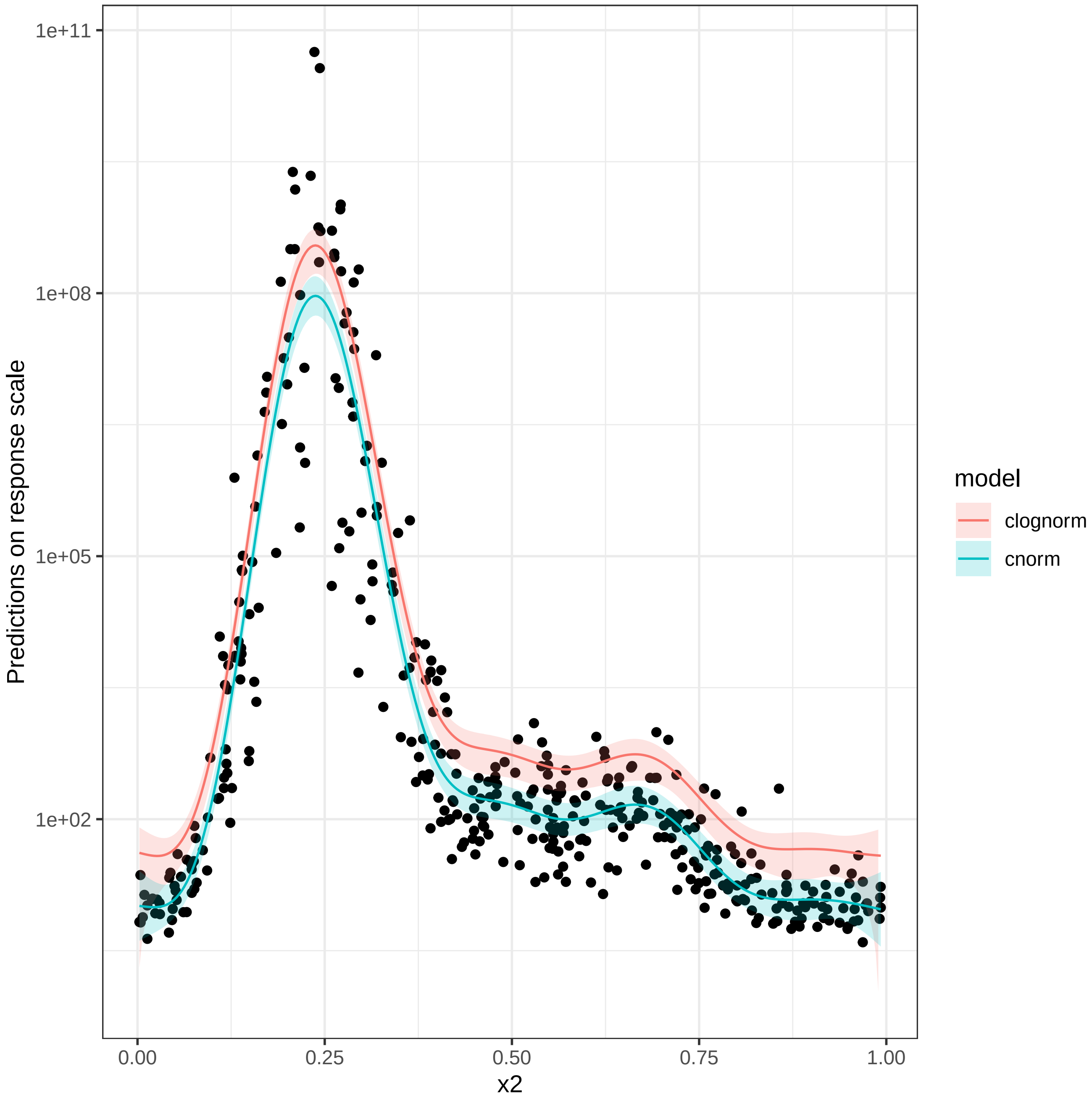

Figure 3: Comparison of corrected and uncorrected predictions.

As usual $L(\theta)$ is the likelihood and $\pi(\theta)$ is the prior on parameter θ. We can write (5) in the following form (equation 2.2 in Tierney et al. (1989)):

$$\mathbb{E}\left[g(\theta)\right] = \frac{\int b_N(\theta)\exp\left[-nh_N(\theta)\right]d\theta}{\int b_D(\theta)\exp\left[-nh_D(\theta)\right]d\theta}, \tag{6}$$

where N and D denote the numerator and denominator versions of the functions, respectively. We can see that our problem fits into the "standard form" where $b_N(\theta) = g(\theta)$, $b_D(\theta) = 1$ and $h_N(\theta) = h_D(\theta) = -(1/n)\log_e\left[L(\theta)\pi(\theta)\right]$ (Tierney and Kadane, 1986), however Tierney et al. (1989) also introduce the "exponential form", which we will use below.

Analysis proceeds by looking at $\mathbb{E}\left\{\exp[sg(\theta)]\right\}$ (the moment generating function). Plugging these values in and taking derivatives will yield Theorem 2 of Tierney et al. (1989) (see paper for fine details on this):

$$\begin{aligned}\mathbb{E}\left[g\left(\hat{\theta}\right)\right] &= g\left(\hat{\theta}\right) + \frac{d}{ds}\log_e \sigma_s|_{s=0} + \frac{d}{ds}\log_e b(\theta_s)|_{s=0} \quad \text{and}\\ \hat{\text{Var}}\left[g\left(\hat{\theta}\right)\right] &= \left[\sigma g'\left(\hat{\theta}\right)\right]^2/n + \frac{d^2}{ds^2}\log_e \sigma_s|_{s=0} + \frac{d^2}{ds^2}\log_e b(\theta_s)|_{s=0}.\end{aligned} \tag{7}$$

This is equivalent to expressions 2e and 2f in Thorson and Kristensen (2016) and equation 6.8 in Thorson and Kristensen (2024). Note that this is only for the case where we have parameter, θ, to estimate. In the case of Tierney et al. (1989), Thorson and Kristensen (2016) and Thorson and Kristensen (2024) the multivariate case is alluded to, but not fully expanded. The rest of the appendix derives this for the specific log-normal case.

Extending to the multivariate case

In the above, we are only looking at univariate θ, we need to extend to the multivariate case. Tierney et al. (1989) state (in the proof of Theorem 2):

"*In the multivariate statement of the theorem, σ_s would be replaced by* $\det\left(\Sigma_s\right)^{1/2}$, *where* $\Sigma_s = \left[D^2 h_s|_{\boldsymbol{\theta}_s}\right]^{-1}$ *and* $\left[\sigma g'(\theta)\right]^2$ *would be replaced by* $(Dg)^T\Sigma(Dg)$, *where* Dg *is the derivative of* g *[with respect to* $\boldsymbol{\theta}$*] evaluated at* $\hat{\boldsymbol{\theta}}$ *and* $\Sigma = \left[D^2 h|_{\hat{\boldsymbol{\theta}}}\right]^{-1}$.*"

Note that here $nh_s(\boldsymbol{\theta}) = nh(\boldsymbol{\theta}) - sg(\boldsymbol{\theta})$.

Doing the suggested substitution and letting $b(\boldsymbol{\theta}) = 1$, (7) becomes

$$\begin{aligned}\mathbb{E}\left[g\left(\hat{\boldsymbol{\theta}}\right)\right] &= g\left(\hat{\boldsymbol{\theta}}\right) + \frac{d}{ds}\log_e \det\left(\Sigma_s\right)^{1/2}|_{s=0} \quad \text{and}\\ \hat{\text{Var}}\left[g\left(\hat{\boldsymbol{\theta}}\right)\right] &= (Dg)^T\Sigma(Dg)/n + \frac{d^2}{ds^2}\log_e \det\left(\Sigma_s\right)^{1/2}\Big|_{s=0}.\end{aligned} \tag{8}$$

Note that the last term in each equation evaluates to zero since $b(\boldsymbol{\theta}) = 1$ (per above) and hence has no derivatives.

Finding $\frac{d\boldsymbol{\theta}_s}{ds}|_{s=0}$

To find quantities such as $\frac{d}{ds}h(\boldsymbol{\theta}_s)$ we need to find $\frac{d\boldsymbol{\theta}_s}{ds}$ (since the resulting derivative is $h'(\boldsymbol{\theta}_s)\frac{d\boldsymbol{\theta}_s}{ds}$, by the chain rule). We will generally evaluate that expression at $s = 0$.

By analogy to equation 3.8 in Tierney et al. (1989), we find

$$\begin{aligned}0 &= \frac{\partial f}{\partial \boldsymbol{\theta}}|_{\boldsymbol{\theta}_s, s} = \frac{\partial}{\partial \boldsymbol{\theta}}\left[nh(\boldsymbol{\theta}) - sg(\boldsymbol{\theta})\right]|_{\boldsymbol{\theta}_s, s}\\ &= n\frac{\partial}{\partial \boldsymbol{\theta}}h(\boldsymbol{\theta}_s) - s\frac{\partial}{\partial \boldsymbol{\theta}}g(\boldsymbol{\theta}_s).\end{aligned}$$

Then taking derivatives wrt s at $s=0$, we have

$$
\begin{aligned}
0 &= \frac{d}{ds}\left[n\frac{\partial}{\partial\boldsymbol{\theta}}h(\boldsymbol{\theta}_s) - s\frac{\partial}{\partial\boldsymbol{\theta}}g(\boldsymbol{\theta}_s)\right]|_{s=0} \\
&= nD^2h(\hat{\boldsymbol{\theta}})\frac{d\boldsymbol{\theta}_s}{ds}|_{s=0} - \frac{\partial}{\partial\boldsymbol{\theta}}g(\hat{\boldsymbol{\theta}}) \\
\frac{d\boldsymbol{\theta}_s}{ds}|_{s=0} &= \frac{1}{n}\left[D^2h(\hat{\boldsymbol{\theta}})\right]^{-1}\frac{\partial}{\partial\boldsymbol{\theta}}g(\hat{\boldsymbol{\theta}}) \\
&= \frac{1}{n}\Sigma\frac{\partial}{\partial\boldsymbol{\theta}}g(\hat{\boldsymbol{\theta}}) \\
&= \frac{1}{n}\Sigma Dg(\hat{\boldsymbol{\theta}}).
\end{aligned}
$$

Finding $\frac{d^2\boldsymbol{\theta}_s}{ds^2}|_{s=0}$

Finding the second derivatives we use the same setup and evaluate

$$
\begin{aligned}
0 &= \frac{d^2}{ds^2}\left[n\frac{\partial}{\partial\boldsymbol{\theta}}h(\boldsymbol{\theta}_s) - s\frac{\partial}{\partial\boldsymbol{\theta}}g(\boldsymbol{\theta}_s)\right]|_{s=0} \\
&= \frac{d}{ds}\left[nD^2h(\boldsymbol{\theta}_s)\frac{d\boldsymbol{\theta}_s}{ds} - \frac{\partial}{\partial\boldsymbol{\theta}}g(\boldsymbol{\theta}_s) - sD^2g(\boldsymbol{\theta}_s)\frac{d\boldsymbol{\theta}_s}{ds}\right]|_{s=0} \\
&= n\frac{d}{ds}D^2h(\boldsymbol{\theta}_s)\frac{d\boldsymbol{\theta}_s}{ds}|_{s=0} - \frac{d}{ds}\frac{\partial}{\partial\boldsymbol{\theta}}g(\boldsymbol{\theta}_s)|_{s=0} - \frac{d}{ds}\left(sD^2g(\boldsymbol{\theta}_s)\frac{d\boldsymbol{\theta}_s}{ds}\right)|_{s=0} \\
&= n\frac{d}{ds}D^2h(\boldsymbol{\theta}_s)\frac{d\boldsymbol{\theta}_s}{ds}|_{s=0} - D^2g(\hat{\boldsymbol{\theta}})\frac{d\boldsymbol{\theta}_s}{ds}|_{s=0} - \left(D^2g(\boldsymbol{\theta}_s)\frac{d\boldsymbol{\theta}_s}{ds}|_{s=0} + \cancelto{0}{s\frac{d}{ds}D^2g(\boldsymbol{\theta}_s)\frac{d\boldsymbol{\theta}_s}{ds}|_{s=0}}\right) \\
&= n\left(\frac{dD^2h(\boldsymbol{\theta}_s)}{ds}|_{s=0}\frac{d\boldsymbol{\theta}_s}{ds} + D^2h(\hat{\boldsymbol{\theta}})\frac{d}{ds}\frac{d\boldsymbol{\theta}_s}{ds}|_{s=0}\right) - D^2g(\hat{\boldsymbol{\theta}})\frac{d\boldsymbol{\theta}_s}{ds}|_{s=0} - D^2g(\boldsymbol{\theta}_s)\frac{d\boldsymbol{\theta}_s}{ds}|_{s=0} \\
&= nD^3h(\hat{\boldsymbol{\theta}})\frac{d\boldsymbol{\theta}_s}{ds}|_{s=0}\frac{d\boldsymbol{\theta}_s}{ds}|_{s=0} + nD^2h(\hat{\boldsymbol{\theta}})\frac{d^2\boldsymbol{\theta}_s}{ds^2}|_{s=0} - 2D^2g(\hat{\boldsymbol{\theta}})\frac{d\boldsymbol{\theta}_s}{ds}|_{s=0} \\
-nD^2h(\hat{\boldsymbol{\theta}})\frac{d^2\boldsymbol{\theta}_s}{ds^2}|_{s=0} &= nD^3h(\hat{\boldsymbol{\theta}})\frac{d\boldsymbol{\theta}_s}{ds}|_{s=0}\frac{d\boldsymbol{\theta}_s}{ds}|_{s=0} - 2D^2g(\hat{\boldsymbol{\theta}})\frac{d\boldsymbol{\theta}_s}{ds}|_{s=0} \\
D^2h(\hat{\boldsymbol{\theta}})\frac{d^2\boldsymbol{\theta}_s}{ds^2}|_{s=0} &= \frac{1}{n}\left(2D^2g(\hat{\boldsymbol{\theta}})\frac{d\boldsymbol{\theta}_s}{ds}|_{s=0} - nD^3h(\hat{\boldsymbol{\theta}})\frac{d\boldsymbol{\theta}_s}{ds}|_{s=0}\frac{d\boldsymbol{\theta}_s}{ds}|_{s=0}\right) \\
\frac{d^2\boldsymbol{\theta}_s}{ds^2}|_{s=0} &= \frac{1}{n}D^2h(\hat{\boldsymbol{\theta}})^{-1}\left(2D^2g(\hat{\boldsymbol{\theta}})\frac{d\boldsymbol{\theta}_s}{ds}|_{s=0} - nD^3h(\hat{\boldsymbol{\theta}})\frac{d\boldsymbol{\theta}_s}{ds}|_{s=0}\frac{d\boldsymbol{\theta}_s}{ds}|_{s=0}\right) \\
&= \frac{1}{n}\Sigma\left(2D^2g(\hat{\boldsymbol{\theta}})\frac{d\boldsymbol{\theta}_s}{ds}|_{s=0} - nD^3h(\hat{\boldsymbol{\theta}})\frac{d\boldsymbol{\theta}_s}{ds}|_{s=0}\frac{d\boldsymbol{\theta}_s}{ds}|_{s=0}\right)
\end{aligned}
$$

Finally substituting $\frac{d\boldsymbol{\theta}_s}{ds}|_{s=0} = \frac{1}{n}\Sigma Dg(\hat{\boldsymbol{\theta}})$ yields:

$$
\begin{aligned}
\frac{d^2\boldsymbol{\theta}_s}{ds^2}|_{s=0} &= \frac{1}{n}\Sigma\left(2D^2g(\hat{\boldsymbol{\theta}})\frac{d\boldsymbol{\theta}_s}{ds}|_{s=0} - nD^3h(\hat{\boldsymbol{\theta}})\frac{d\boldsymbol{\theta}_s}{ds}|_{s=0}\frac{d\boldsymbol{\theta}_s}{ds}|_{s=0}\right) \\
&= \frac{1}{n}\Sigma\left(2D^2g(\hat{\boldsymbol{\theta}})\frac{1}{n}\Sigma Dg(\hat{\boldsymbol{\theta}}) - nD^3h(\hat{\boldsymbol{\theta}})\frac{1}{n}\Sigma Dg(\hat{\boldsymbol{\theta}})\frac{1}{n}\Sigma Dg(\hat{\boldsymbol{\theta}})\right) \\
&= \frac{1}{n^2}\Sigma\left(2D^2g(\hat{\boldsymbol{\theta}})\Sigma Dg(\hat{\boldsymbol{\theta}}) - D^3h(\hat{\boldsymbol{\theta}})\Sigma Dg(\hat{\boldsymbol{\theta}})\Sigma Dg(\hat{\boldsymbol{\theta}})\right)
\end{aligned}
$$

Since$D^3h(\hat{\boldsymbol{\theta}})$ is a 3-tensor (3 dimensional array), we note explicitly that $D^3h(\hat{\boldsymbol{\theta}})\frac{d\boldsymbol{\theta}_s}{ds}|_{s=0}\frac{d\boldsymbol{\theta}_s}{ds}|_{s=0}$ has the contraction

$$
D^3h(\hat{\boldsymbol{\theta}})\Sigma Dg(\hat{\boldsymbol{\theta}})\Sigma Dg(\hat{\boldsymbol{\theta}}) = (\left[D^3h(\hat{\boldsymbol{\theta}})\right]\left[\Sigma Dg(\hat{\boldsymbol{\theta}})\right]_k\left[\Sigma Dg(\hat{\boldsymbol{\theta}})\right]_l .
$$

Finding the derivatives of $\log_e \det(\Sigma_s)^{1/2}$

First derivatives are needed for the expectation:

$$
\begin{aligned}
\frac{d}{ds}\log_e\left[\det(\Sigma_s)^{1/2}\right]|_{s=0} &= \frac{\frac{d}{ds}\det(\Sigma_s)^{1/2}|_{s=0}}{\det(\Sigma)^{1/2}} \\
&= \frac{1/2\det(\Sigma)^{-1/2}\frac{d}{ds}\det\Sigma_s|_{s=0}}{\det(\Sigma)^{1/2}} \\
&= 1/2\det(\Sigma)^{-1}\frac{d}{ds}\det\Sigma_s|_{s=0} \\
&= 1/2\det(\Sigma)^{-1}\det\Sigma\text{trace}\left(\Sigma^{-1}\frac{d}{ds}\Sigma_s|_{s=0}\right) \\
&= 1/2\text{trace}\left(\Sigma^{-1}\frac{d}{ds}\Sigma_s|_{s=0}\right) \\
&= 1/2\text{trace}\left(\Sigma^{-1}\frac{d}{ds}\left[D^2h_s|_{\boldsymbol{\theta}_s}\right]^{-1}|_{s=0}\right) \\
&= -1/2\text{trace}\left(\Sigma^{-1}\left[D^2h_s|_{\boldsymbol{\theta}_s}\right]^{-1}\frac{d}{ds}\left[D^2h(\boldsymbol{\theta})|_{\boldsymbol{\theta}_s}-\frac{s}{n}D^2g(\boldsymbol{\theta})|_{\boldsymbol{\theta}_s}\right]|_{s=0}\left[D^2h_s|_{\boldsymbol{\theta}_s}\right]^{-1}\right) \\
&= -1/2\text{trace}\left(\Sigma^{-1}\Sigma\frac{d}{ds}\left[D^2h(\boldsymbol{\theta})|_{\boldsymbol{\theta}_s}-\frac{s}{n}D^2g(\boldsymbol{\theta})|_{\boldsymbol{\theta}_s}\right]|_{s=0}\Sigma\right) \\
&= -1/2\text{trace}\left(\frac{d}{ds}\left[D^2h(\boldsymbol{\theta})|_{\boldsymbol{\theta}_s}-\frac{s}{n}D^2g(\boldsymbol{\theta})|_{\boldsymbol{\theta}_s}\right]|_{s=0}\Sigma\right) \\
&= -1/2\text{trace}\left(\left[D^3h(\boldsymbol{\theta})\frac{d\boldsymbol{\theta}_s}{ds}-\frac{d}{ds}\left(\frac{s}{n}D^2g(\boldsymbol{\theta})|_{\boldsymbol{\theta}_s}\right)|_{s=0}\right]\Sigma\right) \\
&= -1/2\text{trace}\left(\left[D^3h(\boldsymbol{\theta})\frac{1}{n}Dg(\boldsymbol{\theta})\Sigma-\frac{d}{ds}\left(\frac{s}{n}D^2g(\boldsymbol{\theta})|_{\boldsymbol{\theta}_s}\right)|_{s=0}\right]\Sigma\right) \\
&= -1/2\text{trace}\left(\left[D^3h(\boldsymbol{\theta})\frac{1}{n}Dg(\boldsymbol{\theta})\Sigma-\left(\left(\frac{d}{ds}\frac{s}{n}|_{s=0}\right)D^2g(\boldsymbol{\theta})|_{\boldsymbol{\theta}_s}+\frac{s}{n}\left(\frac{d}{ds}D^2g(\boldsymbol{\theta})|_{s=0}\right)|_{\boldsymbol{\theta}_s}\right)\right]\Sigma\right) \\
&= -1/2\text{trace}\left(\left[D^3h(\boldsymbol{\theta})\frac{1}{n}Dg(\boldsymbol{\theta})\Sigma-\left(\left(\frac{d}{ds}\frac{s}{n}|_{s=0}\right)D^2g(\boldsymbol{\theta})|_{\boldsymbol{\theta}_s}\right)\right]\Sigma\right) \\
&= -1/2\text{trace}\left(\left[\frac{1}{n}D^3h(\boldsymbol{\theta})Dg(\boldsymbol{\theta})\Sigma-\frac{1}{n}D^2g(\boldsymbol{\theta})|_{\boldsymbol{\theta}_s}\right]\Sigma\right) \\
&= \frac{1}{2n}\text{trace}\left(D^2g(\boldsymbol{\theta})|_{\boldsymbol{\theta}_s}\Sigma-D^3h(\boldsymbol{\theta})[\Sigma,\Sigma Dg]\right)
\end{aligned}
$$

This is analogous to the expression given in equation 6.8 in Thorson and Kristensen (2024) and the final expression in section 3.1 in Tierney et al. (1989).

We note that $D^3h(\boldsymbol{\theta})$ forms a 3D tensor (a 3 dimensional array of derivatives). As such, we use the contraction such that $D^3h(\boldsymbol{\theta})[\Sigma,\Sigma Dg] = T_{jkl}\Sigma_{jk}(\Sigma Dg)_l$ where subscripts on the right hand side indicate the indices in the three dimensions j,k,l. If $h(\boldsymbol{\theta})$ were just a normal distribution this term would be zero, as the PDF of a normal distribution does not have third derivatives. However, here we have a log-normal distribution so those derivatives do exist.

For the variance expression, we need to also evaluate $\frac{d^2}{ds^2}\log_e\det(\Sigma_s)^{1/2}$. We begin by writing:

$$A(s) = D^2h(\boldsymbol{\theta}_s) - \frac{s}{n}D^2g(\boldsymbol{\theta}_s),$$

note that then $\Sigma_s = A(s)^{-1}$. Taking the first derivative of $A(s)$ with respect to s we obtain:

$$\begin{aligned}
\frac{dA(s)}{ds} &= \frac{d}{ds} D^2 h(\boldsymbol{\theta}_s) - \frac{d}{ds}\frac{s}{n} D^2 g(\boldsymbol{\theta}_s) \\
&= D^3 h(\boldsymbol{\theta}_s)\frac{d\boldsymbol{\theta}_s}{ds} - \frac{d}{ds}\frac{s}{n} D^2 g(\boldsymbol{\theta}_s) \\
&= D^3 h(\boldsymbol{\theta}_s)\frac{d\boldsymbol{\theta}_s}{ds} - \left(\frac{1}{n} D^2 g(\boldsymbol{\theta}_s) + \frac{s}{n}\frac{d}{ds} D^2 g(\boldsymbol{\theta}_s)\right) \\
&= D^3 h(\boldsymbol{\theta}_s)\frac{d\boldsymbol{\theta}_s}{ds} - \frac{1}{n} D^2 g(\boldsymbol{\theta}_s) - \frac{s}{n} D^3 g(\boldsymbol{\theta}_s)\frac{d\boldsymbol{\theta}_s}{ds}.
\end{aligned}$$

And now the second derivative of $A(s)$ with respect to s and letting $\boldsymbol{\theta}'_s = \frac{d\boldsymbol{\theta}_s}{ds}$ and $\boldsymbol{\theta}''_s = \frac{d^2\boldsymbol{\theta}_s}{ds^2}$:

$$\begin{aligned}
\frac{d^2 A(s)}{ds^2} &= \frac{d}{ds}\left[D^3 h(\boldsymbol{\theta}_s)\boldsymbol{\theta}'_s - \frac{1}{n} D^2 g(\boldsymbol{\theta}_s) - \frac{s}{n} D^3 g(\boldsymbol{\theta}_s)\boldsymbol{\theta}'_s\right] \\
&= \frac{d}{ds}\left(D^3 h(\boldsymbol{\theta}_s)\boldsymbol{\theta}'_s\right) - \frac{d}{ds}\frac{1}{n} D^2 g(\boldsymbol{\theta}_s) - \frac{d}{ds}\frac{s}{n} D^3 g(\boldsymbol{\theta}_s)\boldsymbol{\theta}'_s \\
&= D^4 h(\boldsymbol{\theta}_s)\boldsymbol{\theta}'_s\boldsymbol{\theta}'_s + D^3 h(\boldsymbol{\theta}_s) D^2\boldsymbol{\theta}_s - \frac{1}{n} D^3 g(\boldsymbol{\theta}_s) D\boldsymbol{\theta}_s - \left[\frac{1}{n} D^3 g(\boldsymbol{\theta}_s) D\boldsymbol{\theta}_s + \frac{s}{n}\frac{d}{ds}\left(D^3 g(\boldsymbol{\theta}_s) D\boldsymbol{\theta}_s\right)\right] \\
&= D^4 h(\boldsymbol{\theta}_s)\boldsymbol{\theta}'_s\boldsymbol{\theta}'_s + D^3 h(\boldsymbol{\theta}_s)\boldsymbol{\theta}''_s - \frac{1}{n} D^3 g(\boldsymbol{\theta}_s)\boldsymbol{\theta}'_s - \left[\frac{1}{n} D^3 g(\boldsymbol{\theta}_s)\boldsymbol{\theta}'_s + \frac{s}{n}\left(\frac{d}{ds}\left[D^3 g(\boldsymbol{\theta}_s)\right]\boldsymbol{\theta}'_s + D^3 g(\boldsymbol{\theta}_s)\frac{d}{ds}\boldsymbol{\theta}'_s\right)\right] \\
&= D^4 h(\boldsymbol{\theta}_s)\boldsymbol{\theta}'_s\boldsymbol{\theta}'_s + D^3 h(\boldsymbol{\theta}_s)\boldsymbol{\theta}''_s - \frac{2}{n} D^3 g(\boldsymbol{\theta}_s)\boldsymbol{\theta}'_s - \frac{s}{n}\left(\frac{d}{ds}\left[D^3 g(\boldsymbol{\theta}_s)\right]\boldsymbol{\theta}'_s + D^3 g(\boldsymbol{\theta}_s)\frac{d}{ds}\boldsymbol{\theta}'_s\right).
\end{aligned}$$

We leave the last term with the leading s here as we are going to evaluate the expression at $s = 0$ in a moment so that term will go to zero.

Now note that

$$\frac{1}{2}\log_e \det \Sigma_s = -\frac{1}{2}\log_e \det A(s),$$

so by usual matrix identities

$$\begin{aligned}
\frac{d}{ds}\left[-\frac{1}{2}\log_e \det A(s)\right] &= -\frac{1}{2}\text{trace}\left(A(s)^{-1}\frac{dA(s)}{ds}\right) \quad \text{and} \\
\frac{d^2}{ds^2}\left[-\frac{1}{2}\log_e \det A(s)\right]|_{s=0} &= \frac{1}{2}\text{trace}\left(\left[\Sigma\frac{dA(s)}{ds}|_{s=0}\right]^2\right) - \frac{1}{2}\text{trace}\left(\Sigma\frac{d^2 A}{ds^2}|_{s=0}\right)
\end{aligned}$$

Now taking these pieces:

$$
\begin{aligned}
\frac{d^2}{ds^2}\frac{1}{2}\log_e \det \Sigma_s &= \frac{d^2}{ds^2}\left[-\frac{1}{2}\log_e \det A(s)\right]|_{s=0} \\
&= \frac{1}{2}\text{trace}\left(\left[\Sigma\frac{dA(s)}{ds}|_{s=0}\right]^2\right) - \frac{1}{2}\text{trace}\left(\Sigma\frac{d^2A}{ds^2}|_{s=0}\right) \\
&= \frac{1}{2}\text{trace}\left(\left[\Sigma\left(D^3h(\boldsymbol{\theta}_s)\boldsymbol{\theta}'_s - \frac{1}{n}D^2g(\boldsymbol{\theta}_s) - \cancelto{0}{\frac{s}{n}D^3g(\boldsymbol{\theta}_s)\boldsymbol{\theta}'_s}\right)|_{s=0}\right]^2\right) \\
&\quad - \frac{1}{2}\text{trace}\left(\Sigma\left(D^4h(\boldsymbol{\theta}_s)\boldsymbol{\theta}'_s\boldsymbol{\theta}'_s + D^3h(\boldsymbol{\theta}_s)\boldsymbol{\theta}''_s - \frac{2}{n}D^3g(\boldsymbol{\theta}_s)\boldsymbol{\theta}'_s - \cancelto{0}{\frac{s}{n}\left(\frac{d}{ds}\left[D^3g(\boldsymbol{\theta}_s)\right]\boldsymbol{\theta}'_s + D^3g(\boldsymbol{\theta}_s)\frac{d}{ds}\boldsymbol{\theta}'_s\right)}\right)|_{s=0}\right) \\
&= \frac{1}{2}\text{trace}\left(\left[\Sigma\left(D^3h(\hat{\boldsymbol{\theta}})\boldsymbol{\theta}'_s|_{s=0} - \frac{1}{n}D^2g(\hat{\boldsymbol{\theta}})\right)\right]^2\right) - \frac{1}{2}\text{trace}\Bigg(\Sigma\Bigg(D^4h(\hat{\boldsymbol{\theta}})\boldsymbol{\theta}'_s|_{s=0}\boldsymbol{\theta}'_s|_{s=0} + D^3h(\hat{\boldsymbol{\theta}})\boldsymbol{\theta}''_s|_{s=0} - \frac{2}{n}D^3g(\hat{\boldsymbol{\theta}} \\
&= \frac{1}{2}\text{trace}\left(\left[\Sigma\left(D^3h(\hat{\boldsymbol{\theta}})\boldsymbol{\theta}'_s|_{s=0} - \frac{1}{n}D^2g(\hat{\boldsymbol{\theta}})\right)\right]^2\right) \\
&\quad - \frac{1}{2}\text{trace}\left(\Sigma\left(D^4h(\hat{\boldsymbol{\theta}})\boldsymbol{\theta}'_s|_{s=0}\boldsymbol{\theta}'_s|_{s=0} + D^3h(\hat{\boldsymbol{\theta}})\boldsymbol{\theta}''_s|_{s=0} - \frac{2}{n}D^3g(\hat{\boldsymbol{\theta}})\boldsymbol{\theta}'_s|_{s=0}\right)\right)
\end{aligned}
$$

and substituting the expressions for $\boldsymbol{\theta}'_s$ and $\boldsymbol{\theta}''_s$:

$$
\begin{aligned}
\frac{d^2}{ds^2}\frac{1}{2}\log_e \det \Sigma_s &= \frac{1}{2}\text{trace}\left(\left[\Sigma\left(D^3h(\hat{\boldsymbol{\theta}})\boldsymbol{\theta}'_s|_{s=0} - \frac{1}{n}D^2g(\hat{\boldsymbol{\theta}})\right)\right]^2\right) - \frac{1}{2}\text{trace}\Bigg(\Sigma\Bigg(D^4h(\hat{\boldsymbol{\theta}})\boldsymbol{\theta}'_s|_{s=0}\boldsymbol{\theta}'_s|_{s=0} + D^3h(\hat{\boldsymbol{\theta}})\boldsymbol{\theta}''_s|_{s=0} - \frac{2}{n}D^3g(\hat{\boldsymbol{\theta}} \\
&= \frac{1}{2}\text{trace}\left(\left[\Sigma\left(D^3h(\hat{\boldsymbol{\theta}})\frac{1}{n}\Sigma Dg(\hat{\boldsymbol{\theta}}) - \frac{1}{n}D^2g(\hat{\boldsymbol{\theta}})\right)\right]^2\right) \\
&\quad - \frac{1}{2}\text{trace}\Bigg(\Sigma\Bigg(D^4h(\hat{\boldsymbol{\theta}})\frac{1}{n}\Sigma Dg(\hat{\boldsymbol{\theta}})\frac{1}{n}\Sigma Dg(\hat{\boldsymbol{\theta}}) + D^3h(\hat{\boldsymbol{\theta}})\left[\frac{1}{n^2}\Sigma\left(2D^2g(\hat{\boldsymbol{\theta}})\Sigma Dg(\hat{\boldsymbol{\theta}}) - D^3h(\hat{\boldsymbol{\theta}})\Sigma Dg(\hat{\boldsymbol{\theta}})\Sigma Dg(\hat{\boldsymbol{\theta}})\right)\right] - \frac{2}{n} \\
&= \frac{1}{2n^2}\text{trace}\Bigg(\left[\Sigma\left(D^3h(\hat{\boldsymbol{\theta}})\Sigma Dg(\hat{\boldsymbol{\theta}}) - D^2g(\hat{\boldsymbol{\theta}})\right)\right]^2 \\
&\quad - \Sigma\left(D^4h(\hat{\boldsymbol{\theta}})\Sigma Dg(\hat{\boldsymbol{\theta}})\Sigma Dg(\hat{\boldsymbol{\theta}}) + D^3h(\hat{\boldsymbol{\theta}})\left[\Sigma\left(2D^2g(\hat{\boldsymbol{\theta}})\Sigma Dg(\hat{\boldsymbol{\theta}}) - D^3h(\hat{\boldsymbol{\theta}})\Sigma Dg(\hat{\boldsymbol{\theta}})\Sigma Dg(\hat{\boldsymbol{\theta}})\right)\right] - 2D^3g(\hat{\boldsymbol{\theta}})\Sigma Dg(\hat{\boldsymbol{\theta}})\right)
\end{aligned}
$$

Putting it together

Substituting the above into (8) we then obtain:

$$
\begin{aligned}
\mathbb{E}\left[g\left(\hat{\boldsymbol{\theta}}\right)\right] &= g\left(\hat{\boldsymbol{\theta}}\right) + \frac{1}{2n}\text{trace}\left(D^2g(\boldsymbol{\theta})|_{\boldsymbol{\theta}_s}\Sigma - D^3h(\boldsymbol{\theta})[\Sigma, \Sigma Dg]\right) \quad \text{and} \\
\hat{\text{Var}}\left[g\left(\hat{\boldsymbol{\theta}}\right)\right] &= (Dg)^T\Sigma(Dg)/n + \frac{1}{2n^2}\text{trace}\Bigg(\left[\Sigma\left(D^3h(\hat{\boldsymbol{\theta}})\Sigma Dg(\hat{\boldsymbol{\theta}}) - D^2g(\hat{\boldsymbol{\theta}})\right)\right]^2 \\
&\quad - \Sigma\left(D^4h(\hat{\boldsymbol{\theta}})\Sigma Dg(\hat{\boldsymbol{\theta}})\Sigma Dg(\hat{\boldsymbol{\theta}}) + D^3h(\hat{\boldsymbol{\theta}})\left[\Sigma\left(2D^2g(\hat{\boldsymbol{\theta}})\Sigma Dg(\hat{\boldsymbol{\theta}}) - D^3h(\hat{\boldsymbol{\theta}})\Sigma Dg(\hat{\boldsymbol{\theta}})\Sigma Dg(\hat{\boldsymbol{\theta}})\right)\right] - 2D^3g(\hat{\boldsymbol{\theta}})\Sigma Dg(\hat{\boldsymbol{\theta}})\right)\Bigg).
\end{aligned}
\tag{9}
$$

In practice, we can obtain $\mathbf{V}_{\boldsymbol{\theta}} = \Sigma/n$ as the posterior covariance matrix (via `mgcv::vcov`, see Wood et al. (2016) and Miller (2025) for more information on the meaning of these covariance matrices and how to include uncertainty from smoothing parameters).

We can obtain $D^2 g(\hat{\boldsymbol{\theta}}) = \frac{\partial^2 g(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\Big|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}}$ (i.e., the hessian of $g(\boldsymbol{\theta})$ at the REML-optimal parameters) by numerical derivatives (e.g., `numDeriv::hessian`; Gilbert and Varadhan (2006)) or by analytical expression. Since the analytical expression is faster to compute (since we know that $g(\boldsymbol{\beta}) = b^{X\boldsymbol{\beta}}$), we calculate the j, k^{th} entry in the hessian is given by:

$$
\begin{aligned}
\left[D^2 g(\boldsymbol{\beta})\right]_{jk} = \frac{\partial^2 b^{X\boldsymbol{\beta}}}{\partial \beta_j \partial \beta_k} &= \frac{\partial}{\partial \beta_j} \frac{\partial}{\partial \beta_k} b^{X\boldsymbol{\beta}} \\
&= \frac{\partial}{\partial \beta_j} \left[\left(\frac{\partial}{\partial \beta_k} X\boldsymbol{\beta} \right) b^{X\boldsymbol{\beta}} \log_e b \right] \\
&= \frac{\partial}{\partial \beta_j} \left(\boldsymbol{x}_k^T b^{X\boldsymbol{\beta}} \log_e b \right) \\
&= \boldsymbol{x}_k^T \left(\log_e b \right) \frac{\partial}{\partial \beta_j} b^{X\boldsymbol{\beta}} \\
&= \boldsymbol{x}_k^T \left(\log_e b \right) \boldsymbol{x}_j b^{X\boldsymbol{\beta}} \left(\log_e b \right) \\
&= \left(\log_e b \right)^2 \boldsymbol{x}_k^T \boldsymbol{x}_j b^{X\boldsymbol{\beta}}
\end{aligned}
$$

where b is the base of the response (i.e., we are modelling on the $\log_b$ scale) and $\boldsymbol{x}_j$ and $\boldsymbol{x}_k$ indicate a rows/columns of the design matrix X. By extension (and because we need it for the variance term):

$$
\left[D^3 g(\boldsymbol{\beta})\right]_{ijk} = \frac{\partial^3 b^{X\boldsymbol{\beta}}}{\partial \beta_j \partial \beta_k \partial \beta_l} = \left(\log_e b \right)^3 \left(\boldsymbol{x}_l \times \boldsymbol{x}_k^T \boldsymbol{x}_j \right) b^{X\boldsymbol{\beta}},
$$

where $\left(\boldsymbol{x}_l \times \boldsymbol{x}_k^T \boldsymbol{x}_j \right)$ is an order 3 tensor (i.e., a 3-dimensional array).

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Note that the derivations were checked using Claude Opus 5. A human then checked these calculations.

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